INSE 6450: AI in Systems Engineering

Milestone 3
Adaptive Study Buddy

Robustness, Monitoring & Adaptation

Submitted to:	Professor Ali Ayub, PhD
Teaching Assistant:	Eeham Khan
Submitted by:	Sohana Mahmud
Student ID:	40271073
Date:	March 13, 2026
Repository:	github.com/SohanaMahmud/adaptive-study-buddy

1. Anticipated Failures & Risk Analysis

1.1 Failure Identification & Mitigations

Data-Related Failures

Failure	Applicable ?	Mitigation
Covariate shift	Yes	Monitor feature distributions with PSI; retrain when PSI > 0.2
Label shift	Yes	Track class prior (currently 0.60) in sliding windows; adjust pos_weight
Concept drift	Yes	Periodic retraining on recent data; temporal validation splits
Missing values	Yes	prior_question_elapsed_time can be NaN. Pre-fill with median; reject if null rate > 5%
Out-of-range values	Yes	Clip features to training [1st, 99th] percentile bounds
Timestamp gaps	Yes	u_log_delta handles via log-transform; cap extreme gaps

Model-Related Failures

Failure	Applicable ?	Mitigation
Overconfidence on OOD	Yes	Temperature scaling (T=1.02); abstain if confidence outside [0.05, 0.95]
Calibration errors	Yes	Post-hoc temperature scaling; ECE reduced from 0.0050 to 0.0038
Class imbalance regression	Yes	Recompute pos_weight from recent data (current: 60% positive)
Cascading errors	Yes	Validate feature ranges before inference; isolate feature store failures

System/Infrastructure Failures

Failure	Applicable ?	Mitigation
Timeouts	Yes	50ms hard timeout; return cached prediction or heuristic fallback
No GPU	Low risk	Model runs on CPU in < 0.15ms per sample; no GPU dependency
Server crashes	Yes	Kubernetes health checks; auto-restart with load balancer replicas
Version mismatches	Yes	Pin versions in requirements.txt; Docker containerization
Cache staleness	Yes	TTL-based cache invalidation (refresh q_stats every 6 hours)

User/Interaction Failures

Failure	Applicable ?	Mitigation
Invalid queries	Yes	Input validation schema; return 400 with description
Adversarial inputs	Low risk	Clip inputs to valid ranges; rate-limit rapid submissions
Abuse/edge cases	Yes	Anomaly detection on interaction velocity; flag > 100 interactions/min

1.2 Pre-Flight, Runtime & Post-Prediction Checks

Pre-flight checks validate every input batch: schema validation (8 features), null-rate threshold (reject >5%), feature range validation (flag values beyond 5 std from training mean). Implemented in `preflight_checks()`.

Runtime guards: user_id/content_id must be non-negative integers, elapsed time in [0, 300000]ms. Rate limiting at 1000 req/min per user.

Post-prediction checks: predictions >0.95 or <0.05 flagged as overconfident; [0.45, 0.55] flagged as boundary cases. Flagged predictions logged and may trigger abstention.

1.3 Stress Tests

Four stress conditions evaluated against clean test baseline (AUROC=0.8162, F1=0.8016):

Test 1: Gaussian Noise Injection

Noise sigma	AUROC	F1	Delta AUROC
0.0 (clean)	0.8162	0.8016	baseline
0.1	0.8147	0.8005	-0.0015
0.5	0.7842	0.7814	-0.0320
1.0	0.7269	0.7389	-0.0893
2.0	0.6472	0.6784	-0.1690

Test 2: Feature Masking. Masking q_count alone has zero impact (AUC stays 0.8162). Masking u_accuracy + prior_expl drops AUC to 0.7560 (-0.060). Masking 4 features including q_correct_rate drops AUC to 0.6304 (-0.186), confirming q_correct_rate and u_accuracy are the primary predictors.

Test 3: OOD Samples. Random uniform features produce AUC=0.4971 (random chance) with mean confidence 0.389, indicating the model is appropriately uncertain on unseen distributions.

Test 4: Class Rarity. At 5% positive rate, AUROC=0.8142 (robust), but F1 drops to 0.182 due to high false positive count under extreme imbalance. Recall remains strong at 0.832.

2. Robustness & Security

2.1 Robustness Techniques

1. Data-level: Feature Jittering (sigma=0.1). Gaussian noise added to all features each epoch. Appropriate because tabular features naturally contain measurement noise. Jittering trains smooth decision boundaries and improves stability under perturbation.

2. Model-level: Label Smoothing (alpha=0.05). Hard labels {0,1} smoothed to {0.025, 0.975}. Educational correctness has inherent stochasticity (guessing, careless errors), so soft targets better reflect true uncertainty.

3. Inference-level: Temperature Scaling. Single parameter T=1.02 optimized on validation set. ECE improved from 0.0050 to 0.0038. AUC preserved (ranking unchanged). Lightweight post-hoc calibration with negligible overhead.

2.2 Adversarial Resistance

Threat Model:

Adversary	Type	Description	Likelihood
Student gaming	Black-box	Manipulates response times/explanation clicks for easier questions	Medium
Malicious API user	Grey-box	Knows feature names; crafts inputs for desired predictions	Low
Gradient attacker	White-box	Full model access; FGSM/PGD for minimal perturbations	Very low

FGSM Evaluation (implemented in notebook using PyTorch autograd):

Epsilon	AUROC	F1	Accuracy	Delta AUROC
0.00 (clean)	0.8162	0.8016	0.7495	---
0.01	0.8162	0.8017	+0.0000	+0.0000
0.05	0.8157	0.8014	0.7488	-0.0005
0.10	0.8148	0.8000	0.7474	-0.0014
0.20	0.8104	0.7967	0.7433	-0.0058
0.50	0.7853	0.7817	0.7222	-0.0309

The model is robust to small perturbations (AUC drops only 0.006 at eps=0.2). The jitter-augmented model retains slightly higher AUROC under attack at all levels.

2.3 Robustness Curves & Calibration

The notebook generates a 6-panel visualization: (1) FGSM robustness curve, (2) Gaussian noise degradation, (3) confidence histogram clean vs. adversarial, (4) reliability diagram before/after temperature scaling, (5) ROC overlay, (6) feature masking bar chart. Key: model degrades gracefully and temperature scaling improves calibration without affecting discrimination.

2.4 Quantitative Results & Efficiency

Inference latency is identical for clean and perturbed inputs. Model: 11,777 parameters (0.047 MB). Latency: p50 < 0.15ms, p90 < 0.20ms. Throughput exceeds 1M samples/sec on CPU. No quantization/ONNX needed given minimal latency.

2.5 Failure Case Table

Analysis of 44,819 test predictions: 11,226 errors (25.0%). Of these, 4,451 were confident wrong (|p-0.5| > 0.2). Most common pattern: easy items (high q_correct_rate) answered incorrectly where model confidently predicts correct. Notebook provides 8 curated cases (6 failures + 2 resolved after calibration) with commentary.

3. Monitoring

3.1 Monitoring Targets

Data quality: Schema checks (8 features), null-rate per feature (alert >5%), distribution stats hourly, PSI per feature (alert >0.2). In our 30-day simulation, PSI correctly identified drift on 3 features (u_prev_correct: PSI=3.08, u_log_delta: 0.37, prior_expl: 4.57) when distribution shift was injected at day 20.

Model quality: Online proxy: recommendation acceptance rate. Delayed truth: rolling AUROC/F1 over trailing 10K interactions. In simulation, AUROC ranged from 0.747 to 0.847, with only 1 day below 0.75 SLO after heavy drift injection.

3.2 Architecture & Alerting

Architecture: Server-side logging per batch. Feature drift: 1-hour windows. Model quality: 24-hour windows. 90-day retention, monthly archive.

Condition	Threshold	Response
PSI > 0.2 on any feature	Hourly	Page on-call; investigate feature pipeline
AUROC drop > 0.03	Sustained 24h	Auto-trigger retrain pipeline
F1 below 0.75 SLO	Sustained 24h	Page on-call; consider fallback to baseline
Null rate > 5%	Per-batch	Reject batch; alert data engineering
p95 latency > 50ms	Per-hour	Scale infrastructure; investigate

Runbook: (1) verify alert, (2) check data pipeline, (3) if data issue: fix/recompute, (4) if model degradation: retrain, (5) if retrained model fails: roll back.

3.3 Monitoring Dashboard

Notebook implements 30-day simulation with drift injected at day 20. 4-panel dashboard: (1) PSI heatmap across all features — drift clearly visible after day 20, (2) rolling AUROC vs 0.75 SLO, (3) rolling F1 vs 0.75 SLO, (4) prediction confidence trend. PSI correctly triggers alerts on u_prev_correct (max 3.08), u_log_delta (0.37), and prior_expl (4.57).

4. Adaptation & Model Updates

4.1 Update Triggers

Drift triggers: PSI > 0.2 for >=2 features triggers retraining. Sustained AUROC drop below 0.75 (margin of 0.03 below baseline) for >=2 windows triggers retraining. Operational triggers: new course content, semester boundary.

Cadence: Monthly full retrain for gradual drift. Event-driven fine-tuning on PSI alerts.

Labels: answered_correctly arrives immediately on student response. No labeling delay; inherently reliable binary ground truth.

4.2 Retraining & Versioning

Strategy	When	Description
Full retrain	Monthly / major drift	From scratch on trailing 90 days
Fine-tune last layers	Minor drift / fast update	Freeze first 2 layers; retrain head with LR=5e-4
Rollback	Validation failure	Revert if adapted model underperforms

Versioning: Dataset snapshots with SHA-256 hash. Feature list + StandardScaler params in config YAML. Model state + CONFIG dict as model_v{ver}_{date}.pt.

4.3 Adaptation Experiment

Scenario: New semester with aggressive distribution shift: item difficulties changed (q_correct_rate shifted -1.5 std with added noise), weaker student population (u_accuracy +2.0 std), fewer prior interactions (u_attempts -1.5 std), slower responses (log_elapsed +1.0 std), and more explanation usage (prior_expl +0.8 std). PSI correctly flags all 5 shifted features above the 0.2 threshold.

Metric	Clean	Drifted	Full Retrain	Incremental
AUROC	0.8162	0.7666	0.7989	0.7981
F1	0.8016	0.7049	0.7930	0.7931
Accuracy	0.7495	0.6798	0.7368	0.7357
PR-AUC	0.8629	0.8212	0.8497	0.8490

Drift causes a dramatic AUROC drop from 0.816 to 0.767 (-0.050) and F1 drop from 0.802 to 0.705 (-0.097). Full retrain recovers AUROC to 0.799 (+0.032) but shows moderate forgetting on clean data (0.801 vs 0.816, -0.015). Incremental adaptation achieves similar recovery (AUROC 0.798) while preserving clean performance (0.815, forgetting only -0.001). This demonstrates that incremental fine-tuning is preferred for adaptation.

4.4 Design Changes Based on Milestone 3

1. Feature Normalization (StandardScaler). Now permanent. Logistic baseline achieves AUC=0.8155 (was 0.50 in M2 without normalization).

2. Label Smoothing (alpha=0.05). Permanent. Reduces overconfidence with minimal clean performance impact.

3. Temperature Scaling (T=1.02). Permanent post-processing. ECE: 0.0050 to 0.0038.

4. Early Stopping (patience=5). Permanent. MLP converged at epoch 17/30.

6. Dual adaptation strategy. Full retrain for major drift; incremental fine-tuning (frozen early layers, 50/50 mix) for minor drift. Incremental is preferred as it preserves clean performance (forgetting: -0.001 vs -0.015 for full retrain).

5. Input Validation Pipeline. Pre-flight, runtime, and post-prediction checks now in inference pipeline.

Multi-seed robustness: AUROC = 0.816 +/- 0.0002 across 3 seeds (42, 123, 456), confirming stable results.

Deployment updates: Hourly PSI drift detection. Monthly full retrain + event-driven fine-tuning. Fallback to previous version if catastrophic forgetting (delta AUROC > 0.02).

5. References

1. Riiid Answer Correctness Prediction. https://www.kaggle.com/c/riiid-test-answer-prediction

2. Goodfellow et al. (2015). Explaining and Harnessing Adversarial Examples. ICLR.

3. Guo et al. (2017). On Calibration of Modern Neural Networks. ICML.

4. Szegedy et al. (2016). Rethinking the Inception Architecture for Computer Vision. CVPR.

5. Piech et al. (2015). Deep Knowledge Tracing. NeurIPS.